Classification of Satellite Images

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Date: April 27, 2026

1 Purpose

Digital image processing includes two fundamental tasks: image classification and image segmentation. Although these terms are sometimes used interchangeably, they have different meanings in remote sensing. Image segmentation divides an image into regions or groups of pixels to make it easier to analyze. In contrast, image classification assigns a specific label to each pixel based on its spectral (color) characteristics. The primary goal of these techniques is to transform raw digital numbers—representing reflected electromagnetic energy—into actionable geographic information systems (GIS) data. The purpose of this assignment is to classify a satellite image into five distinct land-cover classes using pixel-based analysis. The classes are **Sea, Tree, Road, Field** and **Sand**.

This project uses pixel-based classification, where each pixel is represented by its RGB values as a three-dimensional feature vector $x = [R, G, B]^T$. The Maximum Likelihood Classification (MLC) algorithm is applied to classify pixels using a probabilistic approach, allowing it to handle variations in natural surface characteristics. The study focuses on how statistical parameters, estimated from a small number of training samples, can be used to model different types of satellite image and produce a complete classified image. To improve the accuracy of Gaussian models, several sampling strategies were evaluated.

2 Method and Algorithm

2.1 Theoretical Framework

This study uses a Gaussian-based Maximum Likelihood classifier to categorize image pixels into predefined land-cover classes. Each class is modeled using a Gaussian (normal) distribution defined by a mean vector μ_i and a covariance matrix Σ_i. The mean vector describes the average color of the class, while the covariance matrix captures the variation and relationships between the RGB components. For a given pixel x, the likelihood is computed using:

$$p(x) = \frac{1}{(2\pi)^{d/2} |\Sigma_i|^{1/2}} \exp\left(-\frac{1}{2}(x - \mu_i)^T \Sigma_i^{-1} (x - \mu_i)\right)$$

where:

x: The pixel vector $[R, G, B]^T$.

μ_i: The mean vector representing the average color signature.

Σ_i: The covariance matrix describing the spread and correlation between bands.

d: The dimensionality ($d = 3$ for RGB).

The exponent term $(x - \mu_i)^T \Sigma_i^{-1} (x - \mu_i)$ represents the Squared Mahalanobis Distance. This probabilistic model allows each pixel to be classified into the class with the highest

likelihood, effectively handling natural variations in color within different types.

3 Experiment

The experiment was conducted using the satellite image, which represents a complex coastal environment with different surface types: **Sea, Tree, Road, Field** and **Sand**.

Figure 1: Experiment Satellite

3.1 Sample Selection

Several sampling strategies were tested to improve classification accuracy. Initially, **Random** sampling was applied, but it introduced high spectral noise due to mixed or unclear pixels. To improve this, a **manual Detailed click-based** approach was used, where carefully selected "pure" pixels were chosen for each class. Finally, a **Coordinate-based** random sampling was adopted. In this approach, specific regions (bounding boxes) were defined for classes such as Tree, Road, and Sea, and 10 pixels were randomly selected within each region.

3.2 Observations

The image shows noticeable spectral confusion between Field and Tree classes due to their similar vegetation characteristics. To reduce misclassification, training samples for the Road class were carefully selected from the center of the road or bridge areas. This helped establish a distinct gray-tone signature, improving separation between Road and surrounding Sand regions.

Figure 2: Output 1
Random Sample

Figure 3: Output 2
Manual Detailed Clicked

Figure 4: Output 3
Coordinate-Based

4 Results and Conclusion

The detailed click-based sampling method produced the smoothest classification results; however, it required more time to manually select pixels for each surface class. In contrast, the coordinate-based sampling approach provided a good balance between efficiency and accuracy, ensuring that the estimated statistical parameters were stable and representative.

The final classified image shows clear separation between Sea and Sand due to their distinct spectral characteristics. Some confusion was observed between Tree and Field classes, which is expected in the RGB color space because of their similar vegetation properties.

By applying the Mahalanobis distance within the Maximum Likelihood framework, the method successfully converts raw RGB pixel data into a structured and meaningful classification map.